1. Methodology (Updated)

Evaluation Metrics

To rigorously assess model performance in both Single-Label and Multi-Label scenarios, we moved beyond simple binary accuracy. The updated analysis evaluates each model against Ground Truth (GT) using four distinct cases to account for partial correctness and hallucinations:

- **Case A (Subset/Precision):** Checks if the model's prediction is a *subset* of the Ground Truth. (Did the model avoid hallucinations? High score = High Precision).
- **Case B (Superset/Recall):** Checks if the Ground Truth is a *subset* of the prediction. (Did the model catch *all* real errors? High score = High Recall).
- **Case C (Overlap):** Checks if there is *any* intersection between prediction and Ground Truth. (Did the model get at least one thing right?).
- **Case D (Jaccard Index):** The strict accuracy metric calculated as `Intersection / Union`. This penalizes both missed errors and hallucinations, providing the most balanced view of performance.

Data Cleaning

All model outputs were standardized using a rigorous cleaning pipeline to resolve inconsistencies (e.g., treating "None", "NaN", and empty strings as a unified "N" label; sorting and deduplicating labels like "J, I" to `{'I', 'J'}`).

2. Results: Consensus & Stability

Consensus Collapse (Single vs. Multi)

A critical finding is the degradation of agreement when models move from restricted (Single) to open-ended (Multi) tasks.

- **Single Mode:** Models achieved a consensus rate of **18.0%** (18/100 questions where all 6 models agreed).
- **Multi Mode:** Consensus collapsed to just **9.0%**, indicating that complexity drives divergence. While models may agree on the *primary* error, they disagree significantly on secondary logic or syntax flaws.

Group Dynamics

- **Closed-Source Stability:** The "Closed" group (GPT-5.2, Claude, Gemini) demonstrated significantly higher stability, agreeing with each other **38.0%** of the time in Single Mode.
- **Open-Source Variance:** The "Open" group (DeepSeek, Qwen, OSS) showed lower cohesion with only **25.0%** agreement.

3. Results: Accuracy Analysis

Leaderboard Summary

GPT-5.2 emerged as the most reliable model across both runs, particularly in complex Multi-Label scenarios.

- In **Multi-Label Run 1**, GPT-5.2 achieved the highest **Jaccard Score (56.8%)** and **Exact Match Rate (24.0%)**.
- **GPT-OSS-120b** followed closely with a 50.8% Jaccard score, showing surprising competitiveness for an open model.
- **DeepSeek-v3.2** lagged in this metric (38.7%), primarily due to "over-flagging" errors that did not exist (hallucinations), which penalized its Jaccard score.

Model "Personality" Profiles (Retention vs. Expansion)

The analysis revealed two distinct behavioral profiles:

1. **Conservative (Precision-Focused): Gemini-2.5-flash** is the most cautious, adding the fewest extra labels (Avg ~2.4 per row) when moving to Multi-Mode. It prioritizes avoiding false positives.
2. **Aggressive (Recall-Focused): Qwen3-Coder** and **DeepSeek-v3.2** adopted a "spray and pray" approach, identifying an average of **4.1 labels per row**. While this ensures they catch most errors (High Case B score), it results in significantly lower overall accuracy due to false positives.

4. Advanced Insights

Error Taxonomy Bias

All models exhibited a strong bias toward identifying **Logical Errors (Category J)**.

- **DeepSeek & Qwen:** Almost exclusively focused on Logical errors, creating a "J-Obsession" spike in the radar charts.
- **GPT-OSS-120b:** Was the unique outlier, showing a distinct bias toward **Syntax/Import Errors (Category I)** that other models ignored.

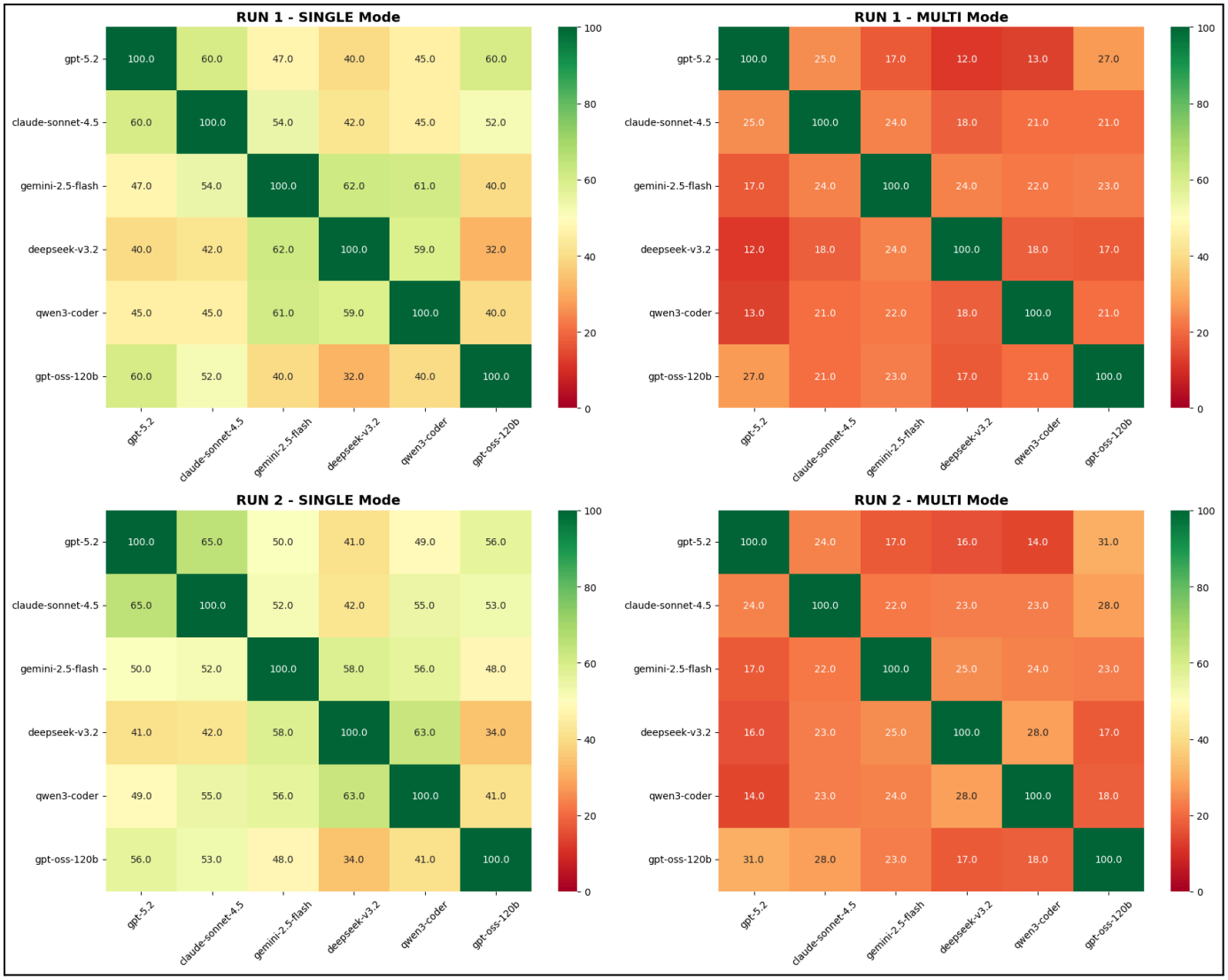
Subjectivity in Reasoning (The "Chaos Score")

For high-level generative tasks (e.g., "Create a LinearEquation class"), the "Chaos Score" reached **5/6**, meaning the 6 models provided **5 different diagnoses** for the same code. This highlights that LLMs currently lack a shared ground truth for complex logic validation.

5. Conclusions

Based on the updated 4-Case analysis, we derive the following key conclusions:

1. **Complexity Breaks Consensus:** Moving from Single to Multi-label classification halves the model agreement rate (18% $\to$ 9%), proving that models diverge wildly when constraints are removed.
2. **GPT-5.2 Remains the Gold Standard:** It consistently leads in Jaccard accuracy (**~57%**), striking the best balance between finding real errors and ignoring fake ones.
3. **The "Personality" Split:** Models are not neutral; they are either **Conservative** (Gemini) or **Aggressive** (Qwen). Users must choose a model based on whether they prioritize *Precision* (avoiding false alarms) or *Recall* (catching every possible bug).
4. **Universal "Logical Error" Bias:** 5 out of 6 models are heavily tuned to find logic flaws (Category J), often hallucinating them in correct code.
5. **Closed Models Offer Stability:** Closed-source models (GPT, Claude, Gemini) are nearly **50% more consistent** as a group than their open-source counterparts.
6. **Subjectivity is High:** In nearly 10% of cases, there is effectively zero agreement among models, suggesting that "automated code review" still requires human oversight for complex reasoning tasks.



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